

Program 10:

step1:

start the minikube with 2 cpus and memory=4096

minikube start --cpus=2 --memory=4096

```
Preparing Kubernetes v1.34.0 on Docker 28.4.0 ...
Verifying Kubernetes components...
  ■ Using image docker.io/kubernetesui/dashboard:v2.7.0
  ■ Using image docker.io/kubernetesui/metrics-scraper:v1.0.8
  ■ Using image gcr.io/k8s-minikube/storage-provisioner:v5
Some dashboard features require the metrics-server addon. To enable all features please run:

    minikube addons enable metrics-server

☀ Enabled addons: storage-provisioner, dashboard, default-storageclass

! /usr/bin/kubectl is version 1.30.14, which may have incompatibilities with Kubernetes 1.34.0.
  ■ Want kubectl v1.34.0? Try 'minikube kubectl -- get pods -A'
Done! kubectl is now configured to use "minikube" cluster and "default" namespace by default
1RV24MC108_SWASTHIK_DEVADIGA@swasthik-Inspiron-15-3515:~/devprg10$ kubectl apply -f deployment.yaml
deployment.apps/nginx-deployment created
1RV24MC108_SWASTHIK_DEVADIGA@swasthik-Inspiron-15-3515:~/devprg10$ kubectl apply -f service.yaml
service/nginx-service created
1RV24MC108_SWASTHIK_DEVADIGA@swasthik-Inspiron-15-3515:~/devprg10$ kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
nginx-deployment-6f9664446b-6rmd7   0/1     ContainerCreating   0          14s
nginx-deployment-6f9664446b-sg2jf   0/1     ContainerCreating   0          14s
1RV24MC108_SWASTHIK_DEVADIGA@swasthik-Inspiron-15-3515:~/devprg10$ kubectl get nodes
NAME      STATUS   ROLES    AGE   VERSION
minikube  Ready    control-plane  34d   v1.34.0
1RV24MC108_SWASTHIK_DEVADIGA@swasthik-Inspiron-15-3515:~/devprg10$ kubectl get svc
NAME      TYPE        CLUSTER-IP   EXTERNAL-IP   PORT(S)          AGE
kubernetes ClusterIP   10.96.0.1     <none>        443/TCP          13d
nginx-service NodePort    10.103.10.4   <none>        80:31385/TCP     29s
1RV24MC108_SWASTHIK_DEVADIGA@swasthik-Inspiron-15-3515:~/devprg10$ kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
nginx-deployment-6f9664446b-6rmd7   1/1     Running    0           63s
nginx-deployment-6f9664446b-sg2jf   1/1     Running    0           63s
1RV24MC108_SWASTHIK_DEVADIGA@swasthik-Inspiron-15-3515:~/devprg10$ kubectl logs nginx-deployment-6f9664446b-6rmd7
/docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt to perform configuration
/docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/
/docker-entrypoint.sh: Launching /docker-entrypoint.d/10-listen-on-ipv6-by-default.sh
10-listen-on-ipv6-by-default.sh: info: Getting the checksum of /etc/nginx/conf.d/default.conf
10-listen-on-ipv6-by-default.sh: info: Enabled listen on IPv6 in /etc/nginx/conf.d/default.conf
/docker-entrypoint.sh: Sourcing /docker-entrypoint.d/15-local-resolvers.envsh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/20-envsubst-on-templates.sh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/30-tune-worker-processes.sh
/docker-entrypoint.sh: Configuration complete; ready for start up
2025/12/30 13:50:55 [notice] 1#1: using the "epoll" event method
2025/12/30 13:50:55 [notice] 1#1: nginx/1.29.4
2025/12/30 13:50:55 [notice] 1#1: built by gcc 14.2.0 (Debian 14.2.0-19)
2025/12/30 13:50:55 [notice] 1#1: OS: Linux 6.8.0-90-generic
2025/12/30 13:50:55 [notice] 1#1: getrlimit(RLIMIT_NOFILE): 1048576:1048576
2025/12/30 13:50:55 [notice] 1#1: start worker processes
2025/12/30 13:50:55 [notice] 1#1: start worker process 29
2025/12/30 13:50:55 [notice] 1#1: start worker process 30
2025/12/30 13:50:55 [notice] 1#1: start worker process 31
2025/12/30 13:50:55 [notice] 1#1: start worker process 32
1RV24MC108_SWASTHIK_DEVADIGA@swasthik-Inspiron-15-3515:~/devprg10$
```

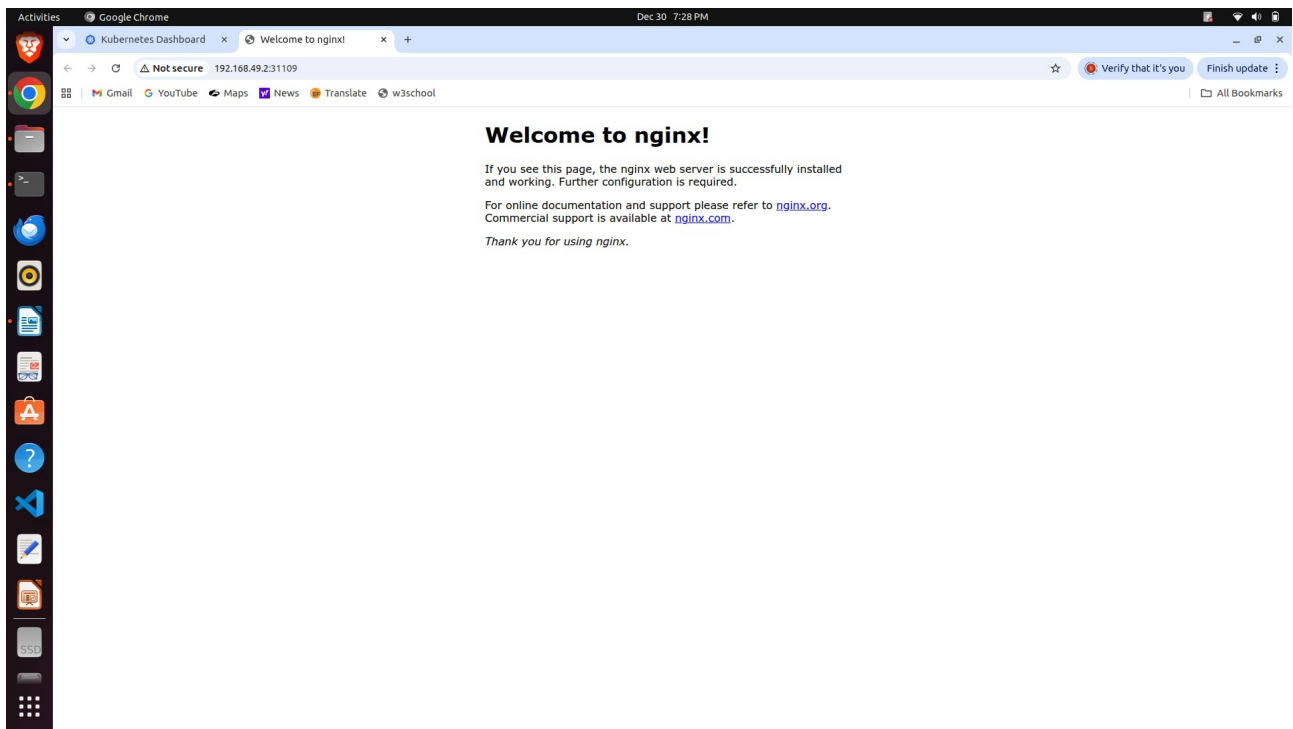
Step2:

create deployment.yaml and service.yaml file

apply two files:

kubectl apply -f deployment.yaml

kubectl apply -f service.yaml



Step3: Monitor the kubernetes Dashboard

minikube dashboard

It will describe :

- 1) pods running, created and number of pods
- 2) services, replica sets, pods details

